Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{3x^2 + 2x - 11}{(x + 2)}$ in the form $mx + c + \frac{r}{(x + 2)}$, and state the value of r.

Final Answer: -3. Polynomial division gives quotient $3x - 4$ with remainder $r = -3$.

2 [Foundational] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{3x^2 + 5x - 16}{(x + 3)}$ in the form $mx + c + \frac{r}{(x + 3)}$, and state the value of r.

Final Answer: -4. Polynomial division gives quotient $3x - 4$ with remainder $r = -4$.

3 [Foundational] • Target: Double Angle Formulas

Express $f(x) = \frac{x^2 + 3x - 7}{(x - 1)}$ in the form $mx + c + \frac{r}{(x - 1)}$, and state the value of r.

Final Answer: -3. Polynomial division gives quotient $x + 4$ with remainder $r = -3$.

4 [Foundational] • Target: Linear Trigonometric Combinations (R-form)

Express $f(x) = \frac{2x^2 - 6x + 9}{(x - 2)}$ in the form $mx + c + \frac{r}{(x - 2)}$, and state the value of r.

Final Answer: 5. Polynomial division gives quotient $2x - 2$ with remainder $r = 5$.

5 [Foundational] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{x^2 + x - 11}{(x + 3)}$ in the form $mx + c + \frac{r}{(x + 3)}$, and state the value of r.

Final Answer: -5. Polynomial division gives quotient $x - 2$ with remainder $r = -5$.

6 [Foundational] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{x^2 - 3x - 4}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.

Final Answer: -4. Polynomial division gives quotient $x - 3$ with remainder $r = -4$.

7 [Foundational] • Target: Double Angle Formulas

Express $f(x) = \frac{3x^2 - 4x + 2}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.

Final Answer: 2. Polynomial division gives quotient $3x - 4$ with remainder $r = 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Express $f(x) = \frac{3x^2 + 4x + 2}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.

Final Answer: 2. Polynomial division gives quotient $3x + 4$ with remainder $r = 2$.

9 [Intermediate] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{3x^2 - 10x + 9}{(x - 2)}$ in the form $mx + c + \frac{r}{(x - 2)}$, and state the value of r.

Final Answer: 1. Polynomial division gives quotient $3x - 4$ with remainder $r = 1$.

10 [Intermediate] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{x^2 + 5x + 8}{(x + 4)}$ in the form $mx + c + \frac{r}{(x + 4)}$, and state the value of r.

Final Answer: 4. Polynomial division gives quotient $x + 1$ with remainder $r = 4$.

11 [Intermediate] • Target: Double Angle Formulas

Express $f(x) = \frac{3x^2 - 9x + 4}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

Final Answer: 4. Polynomial division gives quotient $3x$ with remainder $r = 4$.

12 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Express $f(x) = \frac{x^2 - 3x + 5}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

Final Answer: 5. Polynomial division gives quotient x with remainder $r = 5$.

13 [Intermediate] • Target: Reciprocal Trigonometric Functions

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.

Final Answer: 2. The function is given by $f(x) = x^2 + bx + c$. The minimum value of a quadratic function in the form $ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = 1$, so the minimum occurs at $x = -\frac{b}{2}$. We are given that the minimum occurs at $x = 1$. Therefore, $1 = -\frac{b}{2}$, which implies $b = -2$. Alternatively, we know the vertex is at $(1, -4)$. The vertex form of a quadratic is $f(x) = a(x - h)^2 + k$. Given $a = 1$ and the vertex $(h, k) = (1, -4)$, we have.

14 [Intermediate] • Target: Compound Angle Addition Formulas

Solve $|x + 1| = |2x - 1|$. Give the largest solution.

Final Answer: 2. Working through this functions step: Case 1 : $x + 1 = 2x - 1$. Case 2 : $x + 1 = -(2x - 1)$. Largest solution : $x = 2$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Solve $|x+2| = |2x-3|$. Give the largest solution.

Final Answer: 5. Working through this functions step: Case 1: $x+2 = 2x-3$. Case 2: $x+2 = -(2x-3)$. Largest solution: $x = 5$

16 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Solve $|2x+1| = |x-1|$. Give the largest solution.

Final Answer: 0. Working through this functions step: Case 1: $2x+1 = x-1$. Case 2: $2x+1 = -(x-1)$. Largest solution: $x = 0$

17 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Solve $|x-5| = |2x+5|$. Give the largest solution.

Final Answer: 0. Working through this functions step: Case 1: $x-5 = 2x+5$. Case 2: $x-5 = -(2x+5)$. Largest solution: $x = 0$

18 [Challenging] • Target: Compound Angle Addition Formulas • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Solve $|3x-6| = |-2x|$. Give the largest solution.

Final Answer: 6. Working through this functions step: Case 1: $3x-6 = -2x$. Case 2: $3x-6 = -(-2x)$. Largest solution: $x = 6$

19 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 4.02 The Poisson Distribution

Find $f^{-1}(x)$ if $f(x) = \frac{6}{x}$.

Final Answer: $\frac{6}{x}$. Working through this functions step: $y = \frac{6}{x} \Rightarrow x = \frac{6}{y} \Rightarrow f^{-1}(x) = \frac{6}{x}$ (self − inverse)

20 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 5.01 Further Integration Techniques

Find $f^{-1}(x)$ if $f(x) = \frac{3}{x}$.

Final Answer: $\frac{3}{x}$. Working through this functions step: $y = \frac{3}{x} \Rightarrow x = \frac{3}{y} \Rightarrow f^{-1}(x) = \frac{3}{x}$ (self − inverse)

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal Trigonometric Functions

Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\cot x$

d. $\csc x$

$\sec x - \cos x = \frac{1}{\cos x} - \cos x = \frac{1-\cos^2 x}{\cos x} = \frac{\sin^2 x}{\cos x}$. Dividing by $\tan x = \frac{\sin x}{\cos x}$ gives $\frac{\sin^2 x}{\cos x} \cdot \frac{\cos x}{\sin x} = \sin x$.

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22 [Foundational] • Target: Reciprocal Trigonometric Functions

Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2 ✓

b. -2

c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$. Since $\csc\theta = \frac{1}{\sin\theta}$, $\csc\left(\frac{5\pi}{6}\right) = \frac{1}{1/2} = 2$.

23 [Foundational] • Target: Reciprocal Trigonometric Functions

Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?

a. $1 + \cot^2\theta = \csc^2\theta$ ✓

b. $1 + \csc^2\theta = \cot^2\theta$

c. $\cot^2\theta - \csc^2\theta = 1$

d. $\csc\theta = 1 + \cot\theta$

Dividing $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ yields the fundamental Pythagorean identity $1 + \cot^2\theta = \csc^2\theta$.

24 [Foundational] • Target: Compound Angle Addition Formulas

Use compound angle identities to find the exact value of $\cos(75^\circ) = \cos(45^\circ + 30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

$\cos(A+B) = \cos A\cos B - \sin A\sin B = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$.

25 [Foundational] • Target: Compound Angle Addition Formulas

Simplify $\sin(x+y)\cos y - \cos(x+y)\sin y$.

a. $\sin x$ ✓

b. $\cos x$

c. $\sin(x+2y)$

d. $\cos(x-y)$

Using the subtraction formula $\sin(A-B) = \sin A\cos B - \cos A\sin B$ with $A = x+y$ and $B = y$, this simplifies to $\sin((x+y)-y) = \sin x$.

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26 [Foundational] • Target: Compound Angle Addition Formulas

If $\tan A = 2$ and $\tan B = 3$, what is $\tan(A + B)$?

a. -1 ✓ **b.** 1

c. 5 **d.** $\frac{5}{7}$

$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{2+3}{1-2(3)} = \frac{5}{1-6} = \frac{5}{-5} = -1$.

27 [Foundational] • Target: Double Angle Formulas

If $\sin\theta = \frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$ ✓ **b.** $\frac{24}{25}$

c. $-\frac{7}{25}$ **d.** $\frac{16}{25}$

$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{3}{5}\right)^2 = 1 - 2\left(\frac{9}{25}\right) = 1 - \frac{18}{25} = \frac{7}{25}$.

28 [Foundational] • Target: Double Angle Formulas

Under the same conditions ($\sin\theta = \frac{3}{5}$ acute), what is $\sin(2\theta)$?

a. $\frac{24}{25}$ ✓ **b.** $\frac{7}{25}$

c. $\frac{12}{25}$ **d.** $\frac{6}{5}$

$\cos\theta = \sqrt{1 - 9/25} = \frac{4}{5}$. Then $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Angle Formulas

Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$ ✓ **b.** $\frac{1-\cos(2x)}{2}$

c. $\frac{1+\sin(2x)}{2}$ **d.** $1 - \sin^2 x$

Rearranging $\cos(2x) = 2\cos^2 x - 1$ gives $2\cos^2 x = 1 + \cos(2x) \Rightarrow \cos^2 x = \frac{1+\cos(2x)}{2}$.

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30 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

a. $5\sin(x + 0.927)$ ✓

b. $5\sin(x + 0.643)$

c. $7\sin(x + 0.927)$

d. $25\sin(x + 0.927)$

$R = \sqrt{3^2 + 4^2} = 5$. Expand $R\sin(x+\alpha) = R\cos\alpha\sin x + R\sin\alpha\cos x$. Thus $R\cos\alpha = 3$ and $R\sin\alpha = 4 \Rightarrow \tan\alpha = 4/3 \Rightarrow \alpha = \arctan(4/3) \approx 0.927$ rad.

31 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16 ✓

b. 13

c. 10

d. 20

The combination $5\cos x - 12\sin x$ has amplitude $R = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = 13$. The maximum value is $13 + 3 = 16$.

32 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Solve $\sin(2x) = \cos x$ for $x \in [0, 2\pi)$.

a. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

c. $x = 0, \pi$

d. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$.

33 [Intermediate] • Target: Reciprocal Trigonometric Functions

Find $f^{-1}(x)$ if $f(x) = \frac{4}{x}$.

a. $\frac{4}{x}$ ✓

b. $-\frac{4}{x}$

c. $\frac{1}{4x}$

d. $\frac{x}{4}$

Working through this functions step: $y = \frac{4}{x} \Rightarrow x = \frac{4}{y} \Rightarrow f^{-1}(x) = \frac{4}{x}$ (self − inverse)

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34 **[Intermediate]** • **Target: Compound Angle Addition Formulas**

Find $f^{-1}(x)$ if $f(x) = \frac{5}{x}$.

a. $\frac{1}{5x}$

b. $\frac{x}{5}$

c. $-\frac{5}{x}$

d. $\frac{5}{x}$ ✓

Working through this functions step: $y = \frac{5}{x} \Rightarrow x = \frac{5}{y} \Rightarrow f^{-1}(x) = \frac{5}{x}$ (self − inverse)

35 **[Intermediate]** • **Target: Double Angle Formulas**

Find $f^{-1}(x)$ if $f(x) = \frac{2}{x}$.

a. $\frac{2}{x}$ ✓

b. $\frac{1}{2x}$

c. $\frac{x}{2}$

d. $-\frac{2}{x}$

Working through this functions step: $y = \frac{2}{x} \Rightarrow x = \frac{2}{y} \Rightarrow f^{-1}(x) = \frac{2}{x}$ (self − inverse)

36 **[Intermediate]** • **Target: Linear Trigonometric Combinations (R-form)**

Solve $|3x| = |-x + 4|$. Give the largest solution.

a. 2

b. 0

c. -1

d. **1** ✓

Working through this functions step: Case 1 : $3x = -x + 4$. Case 2 : $3x = -(-x + 4)$. Largest solution : $x = 1$

37 **[Intermediate]** • **Target: Reciprocal Trigonometric Functions**

Find the domain of $f(x) = \sin^{-1}(4x + 2)$.

a. $-\frac{3}{4} \leq x \leq -\frac{1}{4}$ ✓

b. $-\frac{3}{4} < x < -\frac{1}{4}$

c. $-\frac{1}{4} \leq x \leq -\frac{3}{4}$

d. $-1 \leq x \leq 1$

Working through this functions step: Need $-1 \leq 4x + 2 \leq 1 \Rightarrow -\frac{3}{4} \leq x \leq -\frac{1}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Compound Angle Addition Formulas • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Find the domain of $f(x) = \sin^{-1}(6x + 2)$.

a. $-\frac{1}{2} < x < -\frac{1}{6}$

b. $-\frac{1}{2} \leq x \leq -\frac{1}{6}$ ✓

c. $-\frac{1}{6} \leq x \leq -\frac{1}{2}$

d. $-1 \leq x \leq 1$

Working through this functions step: Need $-1 \leq 6x + 2 \leq 1 \Rightarrow -\frac{1}{2} \leq x \leq -\frac{1}{6}$

39 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 4.02 The Poisson Distribution

Find the domain of $f(x) = \sin^{-1}(3x + 3)$.

a. $-\frac{4}{3} \leq x \leq -\frac{2}{3}$ ✓

b. $-\frac{4}{3} < x < -\frac{2}{3}$

c. $-\frac{2}{3} \leq x \leq -\frac{4}{3}$

d. $-1 \leq x \leq 1$

Working through this functions step: Need $-1 \leq 3x + 3 \leq 1 \Rightarrow -\frac{4}{3} \leq x \leq -\frac{2}{3}$

40 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 5.01 Further Integration Techniques

Find the domain of $f(x) = \sin^{-1}(6x - 2)$.

a. $\frac{1}{6} < x < \frac{1}{2}$

b. $-1 \leq x \leq 1$

c. $\frac{1}{2} \leq x \leq \frac{1}{6}$

d. $\frac{1}{6} \leq x \leq \frac{1}{2}$ ✓

Working through this functions step: Need $-1 \leq 6x - 2 \leq 1 \Rightarrow \frac{1}{6} \leq x \leq \frac{1}{2}$

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41 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 5.02 Differential Equations

Find the domain of $f(x) = \sin^{-1}(6x - 3)$.

a. $-1 \le x \le 1$

b. $\frac{1}{3} < x < \frac{2}{3}$

c. $\frac{2}{3} \le x \le \frac{1}{3}$

d. $\frac{1}{3} \le x \le \frac{2}{3}$ ✓

Working through this functions step: Need $-1 \le 6x - 3 \le 1 \Rightarrow \frac{1}{3} \le x \le \frac{2}{3}$

42 [Challenging] • Target: Compound Angle Addition Formulas • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find the domain of $f(x) = \sin^{-1}(4x - 4)$.

a. $\frac{5}{4} \le x \le \frac{3}{4}$

b. $\frac{3}{4} < x < \frac{5}{4}$

c. $-1 \le x \le 1$

d. $\frac{3}{4} \le x \le \frac{5}{4}$ ✓

Working through this functions step: Need $-1 \le 4x - 4 \le 1 \Rightarrow \frac{3}{4} \le x \le \frac{5}{4}$

43 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find the angle between $\mathbf{u} = (2; 6; 9)$ and $\mathbf{v} = (6; -2; 0)$.

a. 90° ✓

b. 104°

c. 72°

d. 69°

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = 0$, so $\cos\theta = 0 \Rightarrow \theta = 90^\circ$

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44 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 6.01 Matrices (AI HL)

Find the angle between $\mathbf{u} = (6; 2; 3)$ and $\mathbf{v} = (2; -6; 0)$.

a. 92°

b. 90° ✓

c. 100°

d. 76°

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = 0,\ \text{so } \cos\theta = 0 \Rightarrow \theta = 90^\circ$

45 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Find the angle between $\mathbf{u} = (3; 4; 12)$ and $\mathbf{v} = (4; -3; 0)$.

a. 92°

b. 77°

c. 102°

d. 90° ✓

Working through this vectors step: $\mathbf{u} \cdot \mathbf{v} = 0,\ \text{so } \cos\theta = 0 \Rightarrow \theta = 90^\circ$

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